



Enterprise Asset Management (EAMS) v15

Asset Deficiencies

End User Documentation

Update 3/1/2024

Asset Deficiencies

MaintStar has an optional (licensed) module called Asset Deficiencies. With this module an organization can record field deficiencies as Assets with the goal of creating a Work Order against said deficiency. An example could be a recurring pothole on a road that needs significant resources to resolve. While submitting a Work Request could suffice, that simply addresses the immediate fix, not resolving the problem. In addition to recording the deficiency as an asset, it can also be connected to GIS so an organization can view the records on a map. Often deficiencies are recorded by Code Enforcement or Inspectors which can then be assigned to work crews or engineers to plan and perform the work.

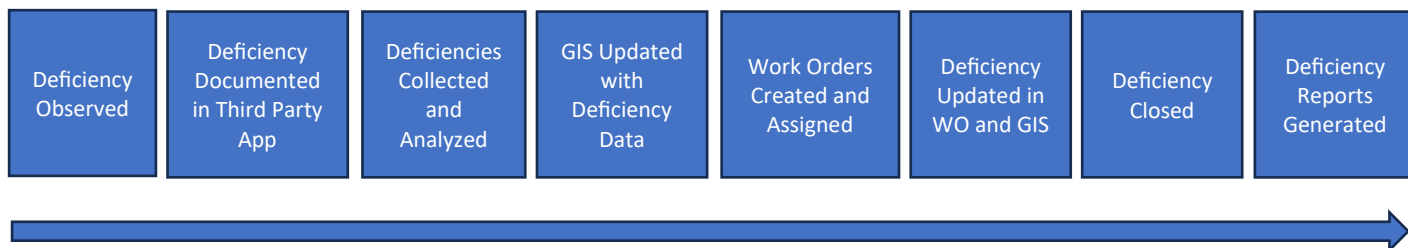
To utilize the Asset Deficiency module:

- It must be licensed
- Deficiency Asset Type(s) must be created or previously created
- Deficiency Types (Activity Codes) must be created or previously created
- Optionally connected to GIS

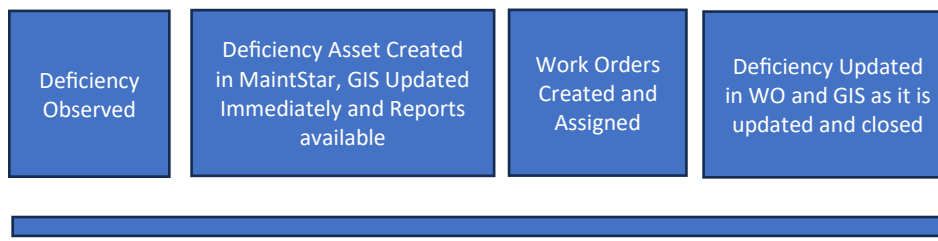
Benefits of using Deficiency Module

Tracking deficiencies has proven to be a challenge for many organizations. Often they will rely on third part tools or applications for tracking. The result is an inefficient method of collecting, analyzing and responding to the needs of the organization. Below are examples of common “As Is” and MaintStar “To Be” workflows.

Typical Workflow:



MaintStar Workflow:



Deficiency Example

The example below shows a recorded deficiency added by Flood Inspectors as an Asset. In addition to Asset Details, pictures are added and it is plotted on a map and uploaded to GIS.

Asset Type: Flood Inspection

Details Attachments Asset Subtype Subsystem Condition Indexes Assessments LTD Meter Update

+ New Asset Type Save Show Asset Count

Asset Type Flood Inspection Workgroup INSPECTION Add Picture

Description Flood Inspection Points

Remarks

Customization Class Flood Inspection Points Parent Asset Type sdChannel - Flood

Reorder Level .000 Reorder Qty .000

Bill Rate .00 Plan Rate

☐ Equipment ☐ Tool ☐ Facility ☐ Vehicle ☐ Motor Pool ☒ Deficiency Asset Type

☒ Active ☐ SVF Flag ☐ Can Borrow / Check Out ☐ Can Calibrate ☐ Linear

Depreciation Formula Downtime %

Cost Share Col Max Dntime Hrs

Default COF Mtg Est Lifespan 0

Geometry Type Point Asset Printout 111 - Inspection Asr

MAINTSTAR Asset Management

Public Works Agency

Full menu

Home (Admin) Work Order Work Request Asset

Asset

Asset Type Asset Group Warranties Barcode Asset Claim Lease Acquisition

INSPECTION

Active Assets (1374)

Search asset... List List+ Tree

1 / 14 by 100

Asset ID	Asset Type
1000	Flood Inspection
1001	Flood Inspection
1002	Flood Inspection
1003	Flood Inspection
1004	Flood Inspection
1005	Flood Inspection
1006	Flood Inspection
1007	Flood Inspection

Asset: 1000 Type: Flood Inspec

Details

New Asset Save

Asset Id 1000

Asset Type Flood Inspection - Fl...

Deficiency Type N42 Concrete Struct

Description 1000

Notes Repair concrete outfall, shows vertical crack, along channel wall. Upstream of railroad.

Add By BA9 - ROBERT BROWN

Add Dt 12/11/2023

Upd By drotninga - AMIE DROT...

Asset

CVS Pharmacy

Colette St

Liviana Dr

Preparing the Environment for Deficiencies

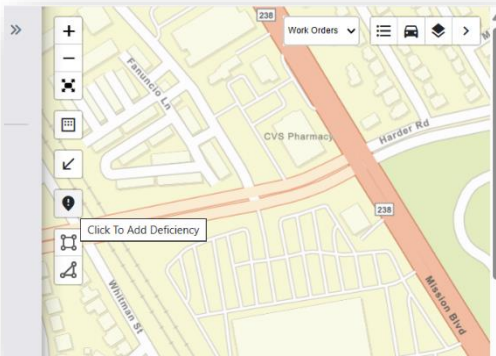
Asset Type: Before the module can be used, it must be set up. As described above, once licensed, an Asset Type must be set up and then GIS (if used).

To set up the Asset Type, navigate to Assets, Asset Type, click New Asset Type. Provide the following base information:

- Asset Type Name
- Asset Type Description
- Asset Type Workgroup
- Active Flag
- Auto Key for Asset ID creation
- Set Geometry Type to Point (for a pin) or Linear (for a line)

Activity Code: In order to classify Deficiencies by type (potholes, inspections, code issue, etc) Activity Codes must be added to the Activity Code table. This table is located in Setup > Tables > Activity Code.

NOTE: Appropriate permissions must be applied to the logged in user to access setup tables.



Asset

Asset Management

Inventory

Reporting

Work Planning

Purchase Orders

Fleet

Setup

Tables

Customer

Vendor

Tables

Table list:

Activity

Work Planning Activity Codes. Was in table acp_work in MS14.

Add

Edit	Activity	Description	Work Unit	Active	Added By
Edit	N81	FLOOD MISCELLANEOUS	LBR HOURS		Convert
Edit	N82	PUMP STATION MISC	LBR HOURS		Convert
Edit	N83	MISCELLANEOUS	LBR HOURS		Convert
Edit	N84	TRAFFIC MISC	LBR HOURS		Convert
Edit	N85	SMALL EQUIP-PM	EQUIP-PM		Convert

GIS Integration: Can be set up to push the recorded deficiencies to an organization’s GIS service. This is helpful to track the locations of multiple deficiencies which in turn can aid in the planning process. Before integration can be completed both MaintStar and the organization’s GIS staff should be engaged. Below is a screenshot of the GIS setup showing the options often selected during the integration.

NOTE: Each environment is different. The below is ONLY an example.

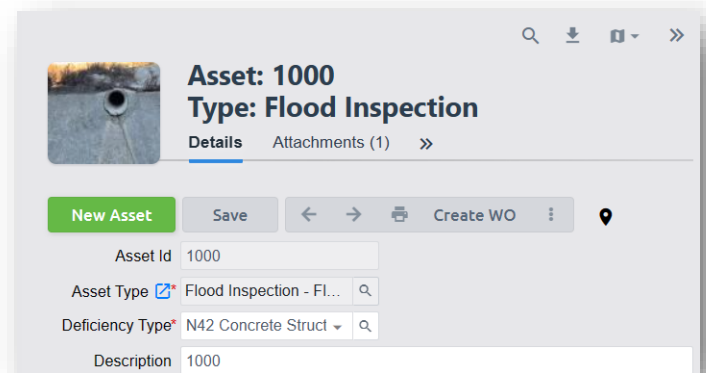
Layer Name	Label	Asset Type	GIS Asset ID Column	GIS Feature Service	Quotes	Min Zoom	Max Zoom	Render	Hide	Top 100	GIS Centric	Is Deficiency	Allow GIS Update	Allow GIS Add	Allow GIS Rename	Allow GIS Delete
Flood Inspection Points		Flood Inspection	DeficiencyID	https://services6.arcgis.com/1X0M0rthS2YWq/Collections/Production/FeatureServer/106		8		json	0	Top 100						

Recording a Deficiency

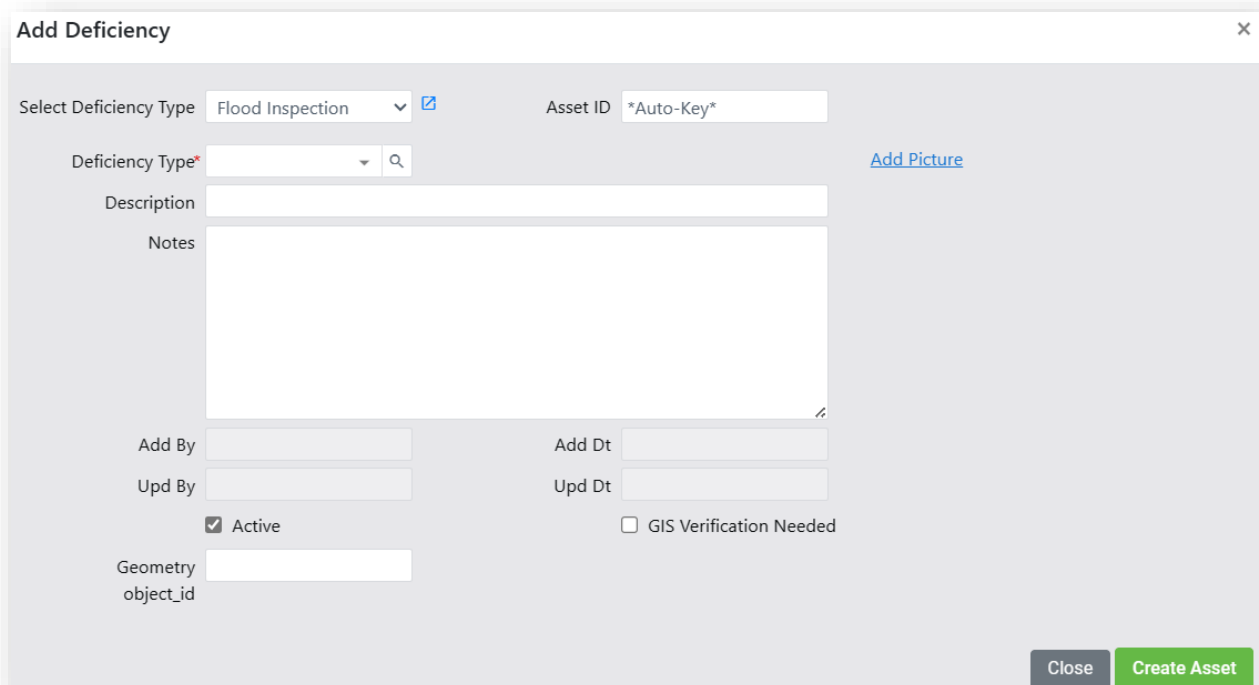
To record a Deficiency, do the following:

- Open Work Order
- Turn on the Map (double headed chevron)

- Locate the Deficiency Marker (black pin) and place on the map
- Once the Marker has been placed a dialog box will appear
- Set the Deficiency Type (Activity Code previously created)
- Provide a Description, Notes and Attachment (if desired)
- Click Create Asset



The screenshot shows the 'Asset: 1000' details page for a 'Flood Inspection'. It includes a search bar, download icon, and a 'Details' tab. Below the tab are buttons for 'New Asset', 'Save', and 'Create WO'. The form fields are: Asset Id (1000), Asset Type (Flood Inspection - Fl...), Deficiency Type (N42 Concrete Struct), and Description (1000).



The 'Add Deficiency' dialog box contains the following fields and controls:

- Select Deficiency Type: Flood Inspection (dropdown with a checkmark icon)
- Asset ID: *Auto-Key* (text field)
- Deficiency Type*: (dropdown with a search icon)
- Description: (text area)
- Notes: (text area)
- Add Picture: (link)
- Add By: (text field)
- Add Dt: (text field)
- Upd By: (text field)
- Upd Dt: (text field)
- ☒ Active
- ☐ GIS Verification Needed
- Geometry object_id: (text field)
- Close (button)
- Create Asset (button)

Creating Work Orders from Deficiencies

The whole point of this effort is to document then assign work from the discovered Deficiencies. Users have a choice of how to create the Work Order: from the map using a selection tool OR from the Deficiency Asset itself.

Creating a Work Order from the Deficiency Asset: This can be used to create Work Orders one by one.

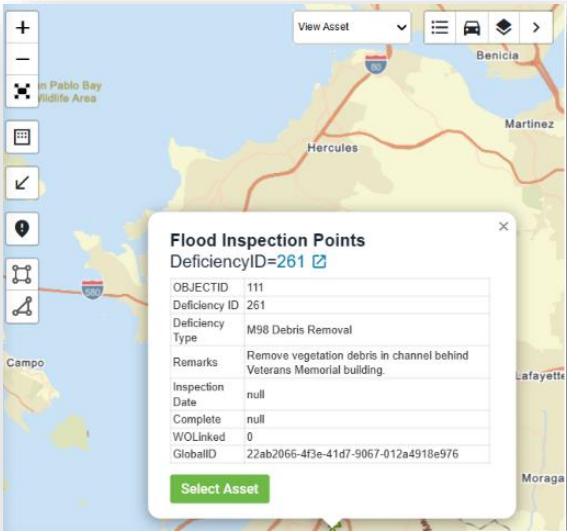
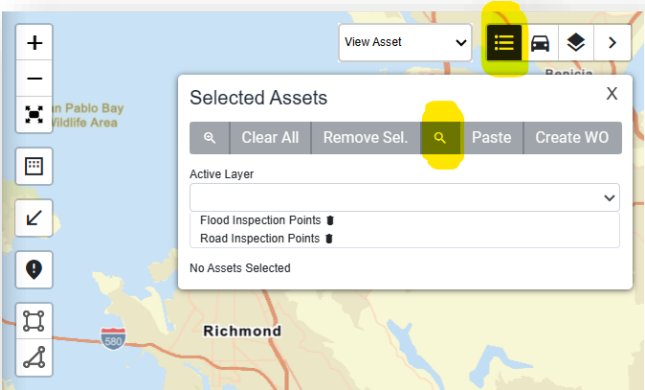
- Click Assets > Assets and locate the Deficiency Assets
- Click the "Create WO" button

- Once the Work Order is created, it will present a confirmation dialog box with the WO Number

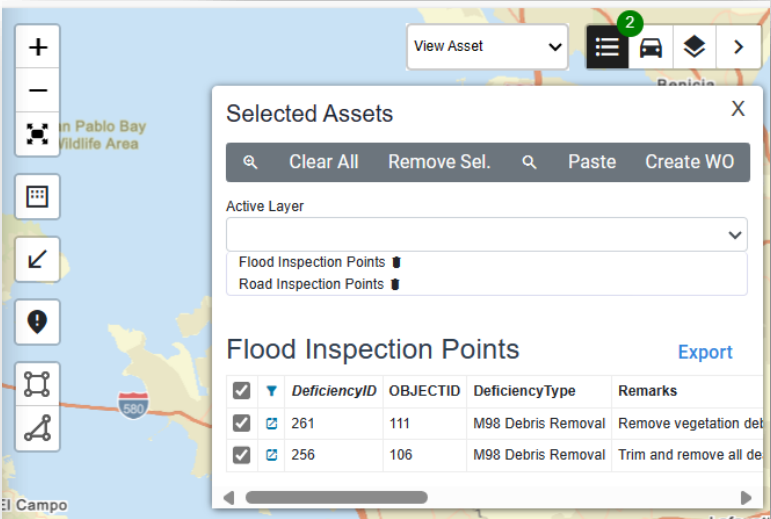
Creating a Work Order from the Map Asset: This can be used to visually locate a particular Deficiency Asset and create a Work Order from the map.

NOTE: to use this capability GIS integration must be set up.

- Click the Map icon to open the map (double headed chevron)
- Make sure the appropriate map layer is identified
- Locate the Deficiency Asset and click the Green “Select Asset” button
- Repeat additional Assets if needed
- All selected Assets will be placed into the selection ‘cart’
- Click the selection ‘cart’ and select all or some of the Assets for which a Work Order is required
- Click the “Create WO” button
- Once the Work Order is created, it will present a confirmation dialog box with the WO Number



Creating a Work Order from the Map Filter: As Asset Deficiencies start to be created, it can be confusing to find the appropriate Deficiency Type for which a Work Order should be created. For instance, the Deficiency Asset Types might contain potholes, debris, graffiti, etc. But if only



“Debris” is to be worked on, then the Map Filter tool can be used to find all of those Deficiencies by their Type.

NOTE: to use this capability GIS integration must be set up.

- Click the Map icon to open the map (double headed chevron)
- Make sure the appropriate map layer is identified
- Click the Select Asset icon (three lines)
- Click the Magnifying Glass
- At this point filter clauses can be applied
- Select the appropriate GIS Layer
- Click Add Clause
- Define a Clause, in this case, Deficiency Type (Activity Code created previously)
- Select “Unique Value” then select the desire value
- Click Run
- All selected Assets will be placed into the selection ‘cart’
- Click the selection ‘cart’ and select all or some of the Assets for which a Work Order is required
- Click the “Create WO” button
- Once the Work Order is created, it will present a confirmation dialog box with the WO Number

